

Sai Ganesh Nelluri

Generative AI Engineer

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PROFESSIONAL SUMMARY

AI/ML and Generative AI Engineer with 5+ years of experience designing and deploying end-to-end machine learning and LLM driven solutions. Expertise in building RAG pipelines, prompt engineering, and multi-agent systems using models like OpenAI GPT, LLaMA, and Azure OpenAI. Proficient in NLP, computer vision, and deep learning with tools like TensorFlow, PyTorch, and Hugging Face. Strong background in MLOps, deploying scalable AI solutions across Azure, AWS, and GCP with CI/CD, Docker, and Kubernetes. Adept at integrating AI into enterprise systems and developing intelligent chat and voice solutions.

SKILLS

Category	Skills
Languages	Python, SQL, Java, Bash, Shell Scripting, R, C#
ML & Deep Learning	PyTorch, TensorFlow, Keras, Scikit-learn, XGBoost, LightGBM, CatBoost, NumPy, Pandas, SciPy, Statsmodels
NLP & Text Processing	spaCy, NLTK, Hugging Face Transformers
Generative AI & LLMs	OpenAI (GPT-3, GPT-3.5, GPT-4, Codex, DALL-E, Whisper, CLIP, ChatGPT), Azure OpenAI, Anthropic Claude (v1-v3), Google Gemini, Meta LLaMA, Mistral, DeepSeek
LLM Frameworks & Tooling	LangChain, LangGraph, LlamaIndex, AutoGen, LangSmith, Transformers, Diffusers, PEFT, Accelerate
Fine-Tuning Techniques	LoRA, QLoRA, Parameter-Efficient Fine-Tuning (PEFT)
Prompt Engineering	Zero-shot, Few-shot, Chain-of-Thought, Function Calling, Custom Prompt Templates
RAG & Vector Databases	Pinecone, FAISS, Weaviate, Chroma, OpenSearch, Adaptive RAG, HyDE, CRAG
MLOps & DevOps	MLflow, Kubeflow, Argo Workflows, Airflow, Docker, Kubernetes, Jenkins, Azure DevOps, DVC
Monitoring & Observability	Prometheus, Grafana, TensorBoard, Azure Monitor, Application Insights
Cloud Platforms	Azure: OpenAI, Cognitive Search, Data Lake, Data Factory, App Service, Container Registry, Key Vault, Active Directory AWS: SageMaker, Bedrock, OpenSearch, S3, DynamoDB, Secrets Manager GCP: Vertex AI
APIs & Backend	FastAPI, Flask, Django, Django REST Framework
UI & Prototyping	Gradio, Streamlit
Conversational AI	Kore.ai, Azure Bot Services, IVR Integrations
Version Control	Git, GitHub, GitLab
Data Visualization	Matplotlib, Seaborn, Plotly

Client: Citi Bank

Role: Data Scientist

May 2026 – Present

Client: Photon

Role: AI Focused Data Engineer

April 2026 – May 2026

Client: Huntington Bank

Role: AI Engineer

May 2024 – March 2026

Responsibilities:

- Designed and deployed **LLM-powered enterprise applications** using **OpenAI (GPT-4o, o1), Anthropic Claude, Google Gemini (Flash), and LLaMA**, supporting chatbots, summarization, and automated decision workflows.
- Engineered advanced **prompting strategies** (zero-shot, few-shot, chain-of-thought, function calling) to improve response relevance, accuracy, and consistency in production.
- Developed **robust ingestion pipelines** for text, embeddings, metadata, and structured data to support scalable retrieval workflows.
- Built **Retrieval-Augmented Generation (RAG)** pipelines using **LangChain, LangGraph, Hugging Face Transformers, and Amazon OpenSearch**, enabling domain-aware and contextual responses.
- Generated high-quality embeddings using **text-embedding-3-large**, powering **semantic search and similarity matching** over large unstructured datasets.
- Designed and managed **Amazon OpenSearch vector indexes**, optimizing shard strategy, recall, and latency for large-scale retrieval.
- Implemented **multi-agent architectures** using **LangChain Agents and LangGraph**, automating complex task orchestration and decision-making.
- Integrated LLM solutions into **enterprise chat and voice platforms**, including **Amazon Lex and IVR systems**, enhancing customer experience through conversational AI.
- Fine-tuned **open-source LLMs (LLaMA, Mistral)** using **LoRA**, deploying optimized inference endpoints on **Amazon SageMaker**.
- Built **scalable backend services** using **Python, FastAPI, and REST APIs**, deployed via **AWS Lambda and API Gateway** for serverless execution.
- Deployed GenAI solutions using **Amazon Bedrock, SageMaker, ECS, and EKS**, supporting containerized and managed model endpoints.
- Implemented **CI/CD pipelines** with **GitHub Actions, Docker, and EKS** to ensure reproducible and reliable GenAI deployments.
- Established **monitoring and observability** using **CloudWatch, Prometheus, Grafana, and LangSmith**, tracking latency, throughput, and model behavior.
- Enforced **enterprise security and compliance** through IAM, KMS encryption, VPC isolation, PII masking, and CloudTrail auditing.
- Evaluated LLM quality using **RAGAS and DeepEval**, assessing faithfulness, relevance, bias, and response quality.

Environment:

Amazon Bedrock, Amazon OpenSearch (AOSS), Amazon SageMaker, Amazon Kendra, LangChain, LangGraph, Hugging Face, FastAPI, AWS Lambda, EKS, ECS, Docker, GitHub Actions, RAGAS, DeepEval, LangSmith, IAM, Cognito, PII Masking

Central Michigan University, Academic Work

Feb 2023 – May 2024

Responsibilities:

- Designed and deployed **end-to-end ML pipelines** on **AWS SageMaker** for classification, prediction, and recommendation systems.
- Built and productionized **NLP solutions** for text classification, sentiment analysis, NER, topic modeling, and summarization using **spaCy, NLTK, and Hugging Face Transformers**.
- Developed **data preprocessing and feature engineering pipelines**, including tokenization, embeddings, and normalization.
- Built and optimized **deep learning models** using **TensorFlow and PyTorch**, addressing challenges such as data imbalance, drift, and scalability.
- Exposed ML models via **FastAPI-based REST services**, deployed using **AWS Lambda and API Gateway**

- Implemented **real-time and batch data pipelines** using **AWS Glue, Kinesis Data Streams, and Lambda**.
- Orchestrated **async ML workflows** using **Celery and Redis** within Django-based systems.
- Established **MLOps workflows** with **MLflow, SageMaker Model Monitor, and CodePipeline**, enabling versioning, monitoring, and automated retraining.
- Monitored **model and data drift**, triggering alerts and retraining to maintain production accuracy.
- Containerized ML services using **Docker** and deployed to **Amazon EKS** for scalable production workloads.
- Collaborated with cross-functional teams to deliver **secure, compliant AI solutions** aligned with enterprise data governance.
- Performed **model evaluation and tuning** using precision, recall, F1, ROC-AUC, Optuna, and SageMaker Automatic Model Tuning.

Environment:

AWS SageMaker, AWS Lambda, AWS Glue, AWS Kinesis, Amazon EKS, Docker, FastAPI, TensorFlow, PyTorch, Hugging Face, MLflow, Optuna, CloudWatch, Prometheus, Grafana

Accenture – Hyderabad, India

Role: Data Scientist

Jul 2020 – Aug 2022

Responsibilities:

- Built and deployed **ML solutions** for classification, regression, clustering, and conversational AI using **Scikit-learn, XGBoost, TensorFlow, and Transformers**.
- Implemented data preprocessing, cleansing, and validation pipelines **using Python and SQL to ensure high-quality inputs for machine learning models**.
- **Designed** feature stores and reusable feature pipelines, **enabling consistent feature usage across training and inference workflows**.
- **Worked extensively with** relational databases, **writing complex SQL queries, joins, window functions, and aggregations to support ML analytics and reporting**.
- **Assisted in** model selection and experimentation **across supervised and unsupervised learning techniques, including clustering and anomaly detection**.
- **Supported** production model troubleshooting, **analyzing prediction errors, data anomalies, and performance degradation**.
- Conducted **EDA and feature engineering** using **Pandas, NumPy, and Matplotlib** to improve model accuracy.
- Developed **secure backend APIs** using **Django, Django REST Framework, Flask, and FastAPI** for ML-powered applications.
- Implemented **role-based access control and authentication** for AI platforms.
- Built **ETL and real-time data pipelines** using **Apache Spark, Hadoop, and Databricks**.
- Developed and deployed **NLP models** for sentiment analysis, NER, and text classification.
- Performed **model validation and hyperparameter tuning** using Grid Search and Bayesian optimization.
- Created **dashboards and reports** using **Power BI and Plotly** for business stakeholders.
- Ensured **code quality** using unit tests and PyTest.
- Used **SQL** extensively for large-scale data analysis and feature preparation.
- Documented workflows to ensure **reproducibility, governance, and compliance**.

Environment:

AWS SageMaker, AWS Lambda, Spark, Databricks, Docker, FastAPI, Django, TensorFlow, PyTorch, Hugging Face, MLflow, CloudWatch, Prometheus, Grafana

Education

- Masters degree from Central Michigan University, Michigan, United States

Dec 2024